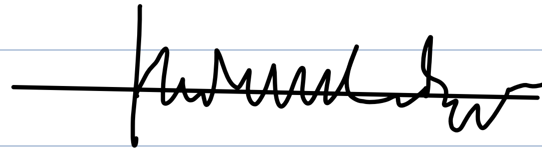
Recall :



Moving Average $\quad X_t = \frac{1}{3}(W_{t-1} + W_t + W_{t+1})$

$$\mu(t) = 0$$

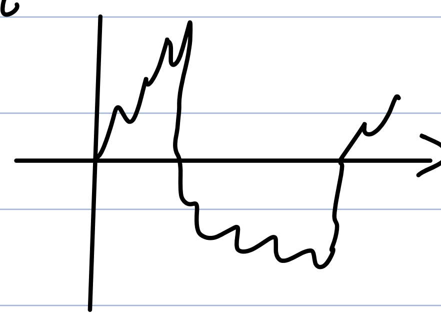
$$\gamma(s,t) = \begin{cases} 0 & |s-t| \geq 3 \\ \frac{1}{9}\sigma_w^2 & |s-t| = 2 \\ \frac{2}{9}\sigma_w^2 & |s-t| = 1 \\ \frac{3}{9}\sigma_w^2 & |s-t| = 0 \end{cases}$$

only relative location matters

vs .

Random Walk $\quad X_t = W_1 + \cdot + W_t$

$$\mu(t) = 0$$

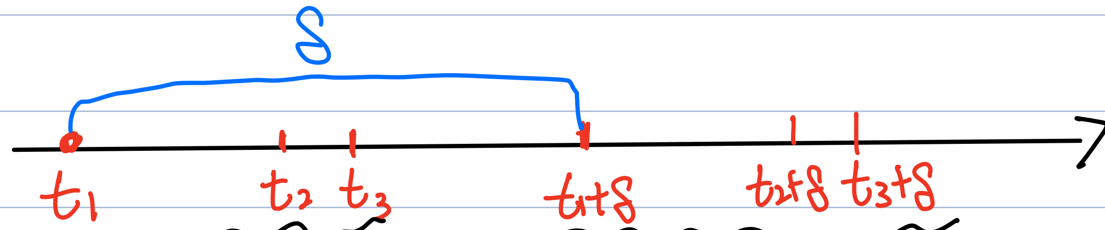


$$\gamma(s,t) = \sigma_w^2 \min(s,t)$$

abs. location matters.

Def.* (strict stationarity)

A time series $\{X_t\}$ is said to be strictly stationary, if the joint distribution of $(X_{t_1}, X_{t_2}, \cdots, X_{t_k})$ is identical to that of $(X_{t_1+S}, X_{t_2+S}, \cdots X_{t_k+S})$, for any time points $t_1, \cdots, t_k$, any k, and any time shift δ.



joint distr. = joint distribution

Remark: ① the def. can be summarized as: all the finite-dimensional joint distributions are invariant under time shift.

② Make sure to distinguish the concept of strict stationarity from: "marginal distributions of $\{X_t\}$ are identical".

imply

Examples of nonstationary time series.

① Signal + noise

$X_t = S_t + V_t$

S_t: trigonometric curve

V_t: Gaussian noise.

It is NOT stationary because

$E[X_t] = S_t$ not a constant.

↳ marginal distributions of X_t's are changing.

② Noise with increasing variance

$X_t = \frac{t}{c} W_t$

some constant

Since variance is changing

↳ marginal distributions are changing.

③ Random Walk

recall: $Var(X_t) = \sigma_\omega^2 t$. changing.

strict stationarity

$\Rightarrow \begin{cases} \text{identical marginal distr.} \Rightarrow \mu(t) = \text{constant.} \\ (X_s, X_t) \overset{d}{=} (X_{s+\delta}, X_{t+\delta}) \Rightarrow Cov(X_s, X_t) = Cov(X_{s+\delta}, X_{t+\delta}) \end{cases}$

has the same distribution

$\Leftrightarrow \gamma(s,t) = \gamma(s+\delta, t+\delta)$

- In your textbook
"stationarity" = "weak stationarity".

- Univariate autocovariance function

(under weak stationarity)

$$\gamma(h) = \gamma(\overset{s}{t+h}, \overset{t}{t})$$

$$\left(\gamma(s,t) = \gamma(s+\delta, t+\delta) \right)$$

$\delta = -t$

$$= \gamma(t+h-t,\ t-t) = \gamma(h,0)$$

Univariate autocorrelation function

$$\rho(h) = \text{Corr}(X_{t+h}, X_t) \overset{?}{=} \frac{\gamma(h)}{\gamma(0)}$$

$$= \text{Corr}(X_h, X_0)$$

$$= \frac{\gamma(h,0)}{\sqrt{\gamma(h,h)\cdot\gamma(0,0)}}$$

$\gamma(h,h) = \gamma(0,0)$
$= \gamma(0) =$ marginal var.

Another important property:

$$\gamma(h) = \gamma(-h)$$

proof:

$$\gamma(h) = \text{Cov}(X_h, X_0)$$

$$\gamma(-h) = \text{Cov}(X_{-h}, X_0)$$

$$= \text{Cov}(X_{-h+h},\ X_{0+h})$$

weak stationarity

$$= Cov(X_0, X_h)$$